

Assignment3

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2026-03-01

```
#preparing data
```

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
set.seed(123)
mydata <- read.csv("UniversalBank-1.csv")
bank <- mydata[,c("Online", "CreditCard", "Personal.Loan")]
colnames(bank) <- c("Online", "CC", "PL")
```

```
#make a partition into training (60%)
```

```
index_train <- createDataPartition(bank$PL, p=0.6, list= FALSE)
train <- bank[index_train,]
valid <- bank[-index_train,]
```

```
#Part A
```

```
pivot <- table(cc=train$CC, pl=train$PL, online=train$Online)
```

```
#part B
```

```
prob.b <- pivot["1", "1", "1"] / (pivot["1", "1", "1"] + pivot["1", "0", "1"])
prob.b
```

```
## [1] 0.1071429
```

```
#part c
```

```
pivot1 <- table(pl=train$PL, online=train$Online)
pivot2 <- table(pl=train$PL, cc=train$CC)
```

```
#part D
```

```

#i
prob_cc <- pivot2["1","1"]/sum(pivot2["1",])
#ii
prob_online <- pivot1["1","1"]/sum(pivot1["1",])
#iii
prob_pl <- sum(train$PL==1)/nrow(train)
#iv
prob_cc_pl0 <- pivot2["0","1"]/sum(pivot2["0",])
#v
prob_online_0 <- pivot1["0","1"]/sum(pivot1["0",])
#vi
prob_pl0 <- sum(train$PL==0)/nrow(train)

```

#part E

```

prob_E <- prob_cc*prob_online*prob_pl/((prob_cc*prob_online*prob_pl)+(prob_cc_pl0*prob_online_0*prob_pl0))
prob_E

```

```
## [1] 0.1105637
```

#part F

```

comparison <- data.frame(method=c("pivot table","Naive Bayes"),
                           probability=c(prob.b,prob_E))
comparison

```

```

##           method probability
## 1 pivot table    0.1071429
## 2 Naive Bayes    0.1105637

```

#The pivot table probability(0.1071) and the Naive Bayes probability(0.1106) are very close. The small difference happens because Naive Bayes assumes CC and Online are independent given PL. The pivot table value is the more accurate estimate since it uses the actual counts.

#part G

```
library(e1071)
```

```

##
## Attaching package: 'e1071'

```

```
## The following object is masked from 'package:ggplot2':
```

```

##
##      element

```

```

train$PL <- as.factor(train$PL)
train$Online <- as.factor(train$Online)
train$CC <- as.factor(train$CC)
nb <- naiveBayes(PL ~ CC + Online, data = train)
customer <- data.frame(CC=factor(1,levels = levels(train$CC)),Online=factor(1,levels = levels(train$Online)))
prediction <- predict(nb,customer,type = "raw")
prob_loan1 <- prediction["1"]
prob_loan1

```

```
##          1
## 0.1105637
```

#I trained a Naive Bayes model using CC and Online to predict PL. For a customer with $CC = 1$ and $Online = 1$, the predicted probability of accepting the loan was 0.1106, which matches the manual Naive Bayes calculation from Part E.